PHYSICAL CUMULATIVE EXAM

October 5, 1996

1.) In group theory, what is a *matrix representation* of a point group, what is the *character* of a representation, how are *classes of conjugate elements* defined, what are *equivalent representations*, and what are *reducible and irreducible representations*?

2.) Consider pointgroup C_{2v}. Assume a Cartesian coordinate system with x,y,z axes, in which the z-axis is along the C_2 axis, and the (z,x)-plane is the paper plane.

a) Determine to which irreducible representations of point group C_{2v} the translations along the x-, y-, and z-axes belong.

b) If a mass m moves with momentum $\mathbf{p} = m\mathbf{v}$ relative to a fixed point O, then the angular momentum $\mathbf{L}$ of the mass with respect to the point O is the vector product $\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}$, where $\mathbf{r}$ is the radius vector from O to the position of the particle. The components of $\mathbf{L}$ in the Cartesian coordinate system are defined as follows:

$$L_x = y\,p_z - z\,p_y$$
$$L_y = z\,p_x - x\,p_z$$
$$L_z = x\,p_y - y\,p_x$$

Determine to which irreducible representations of point group C_{2v} the components L_x, L_y, and L_z belong.

$\mathscr{C}_{2v}$	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$
A_1	1	1	1	1
A_2	1	1	−1	−1
B_1	1	−1	1	−1
B_2	1	−1	−1	1